A
Project Presentation
On
Feature Extraction and Price Prediction for Mobile Phones

Submitted to

NEXT HIKES
IT SOLUTIONS

Submitted by
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The project aims to identify what drives mobile phone prices and how predictive analytics can support business decisions.

The objectives of the project are as follows:

1. To understanding mobile phone price prediction using machine learning and feature analysis.

2. To establishes the business relevance of identifying the factors that influence mobile pricing.

3. To Positions the analysis as a data-driven solution for pricing strategy.

Dataset Information Table

Column Name	Data Type	Non-Null Count
Unnamed: 0	int64	541
Model	object	541
Colour	object	541
Memory	int64	541
RAM	int64	541
Battery_	int64	541
Rear Camera	object	541
Front Camera	object	541
AI Lens	int64	541
Mobile Height	float64	541
Processor_	object	541
Prize	object	541

Table 1: Dataset information table for price prediction of mobile phones

- Provides an overview of dataset structure and variable availability.
- Demonstrates that multiple technical specifications were included.
- Indicates that the dataset contains relevant variables needed for pricing prediction.
- A diverse dataset increases prediction reliability.
- Rich feature availability allows deeper understanding of customer valuation drivers.

Statistical Summary Table of Mobile Dataset

	Unnamed: 0	Memory	RAM	Battery_	AI Lens	Mobile Height
count	541.0	541.0	541.0	541.0	541.0	541.0
mean	289.71	110.55	5.4	4871.59	0.06	16.43
std	182.36	60.6	1.98	780.15	0.24	2.52
min	0.0	16.0	2.0	800.0	0.0	4.5
25%	135.0	64.0	4.0	5000.0	0.0	16.51
50%	273.0	128.0	6.0	5000.0	0.0	16.71
75%	434.0	128.0	8.0	5000.0	0.0	16.94
max	642.0	256.0	8.0	7000.0	1.0	41.94

Table 2: Statistical summary of price prediction for mobile phones

- ❑ Shows mean, minimum, maximum, and distribution range of variables.
- ❑ Highlights the diversity of mobile configurations in the market.
- ❑ Helps identify pricing spread between budget and premium devices.
- ❑ Large variation in mobile specifications supports market segmentation.
- ❑ Organizations can identify product gaps within pricing categories.

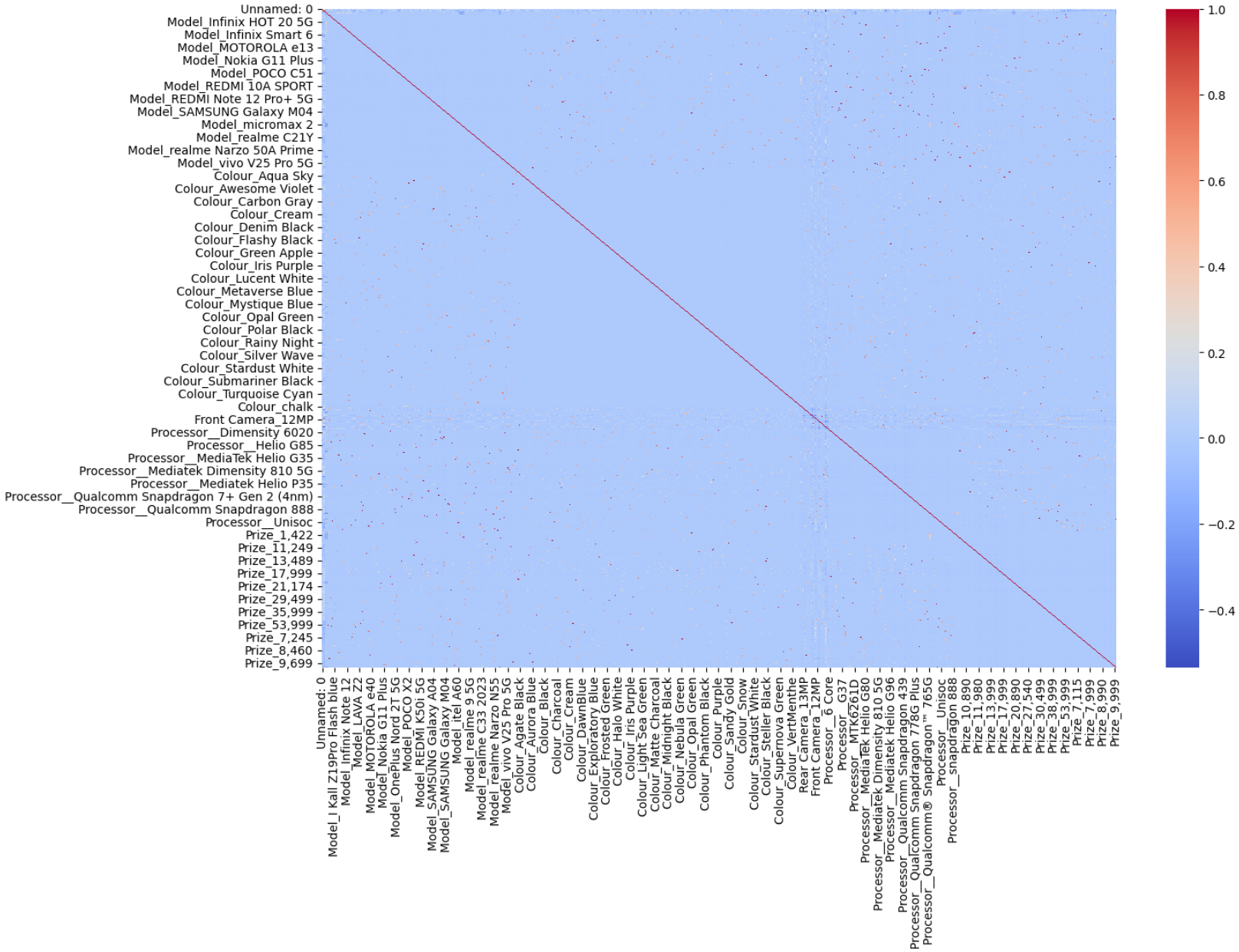


Figure 1: Correlation heatmap of different parameters of mobile phones

- Displays relationships among mobile specifications.
- Strong positive correlations suggest some features increase together.
- High-end smartphones often bundle premium features.
- RAM and Memory likely show strong positive association.
- Battery and camera quality may correlate with price.
- Premium phones combine multiple performance attributes.
- Smartphone pricing is influenced by combined feature ecosystems rather than isolated specifications.

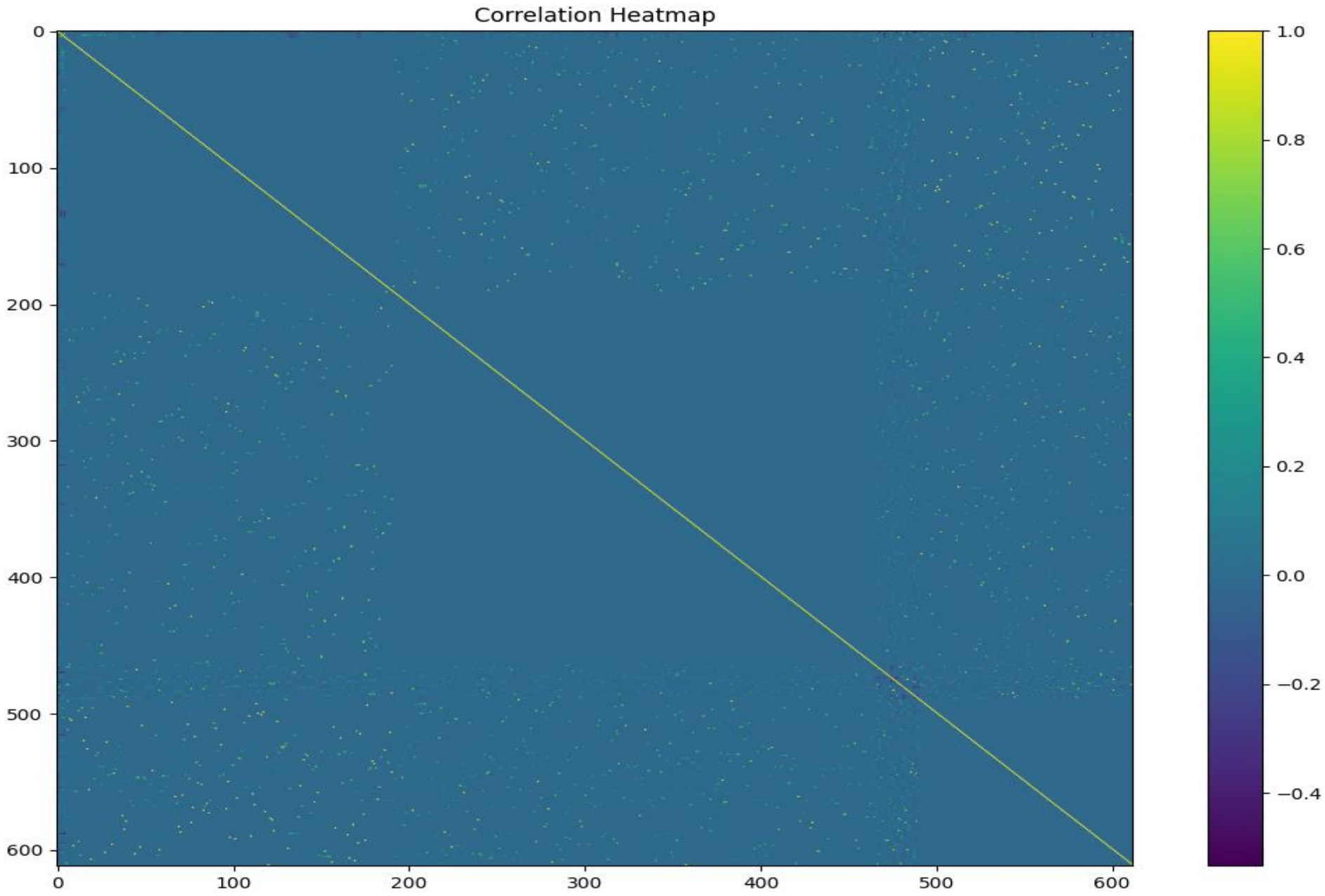


Figure 2: Correlation analysis of price prediction for mobile phones

- Shows which features are most strongly related to mobile price.
- Helps distinguish high-impact vs low-impact variables.
- Technical performance features contribute more strongly to price.
- Cosmetic dimensions have lower influence.
- Organizations should prioritize features customers truly value.

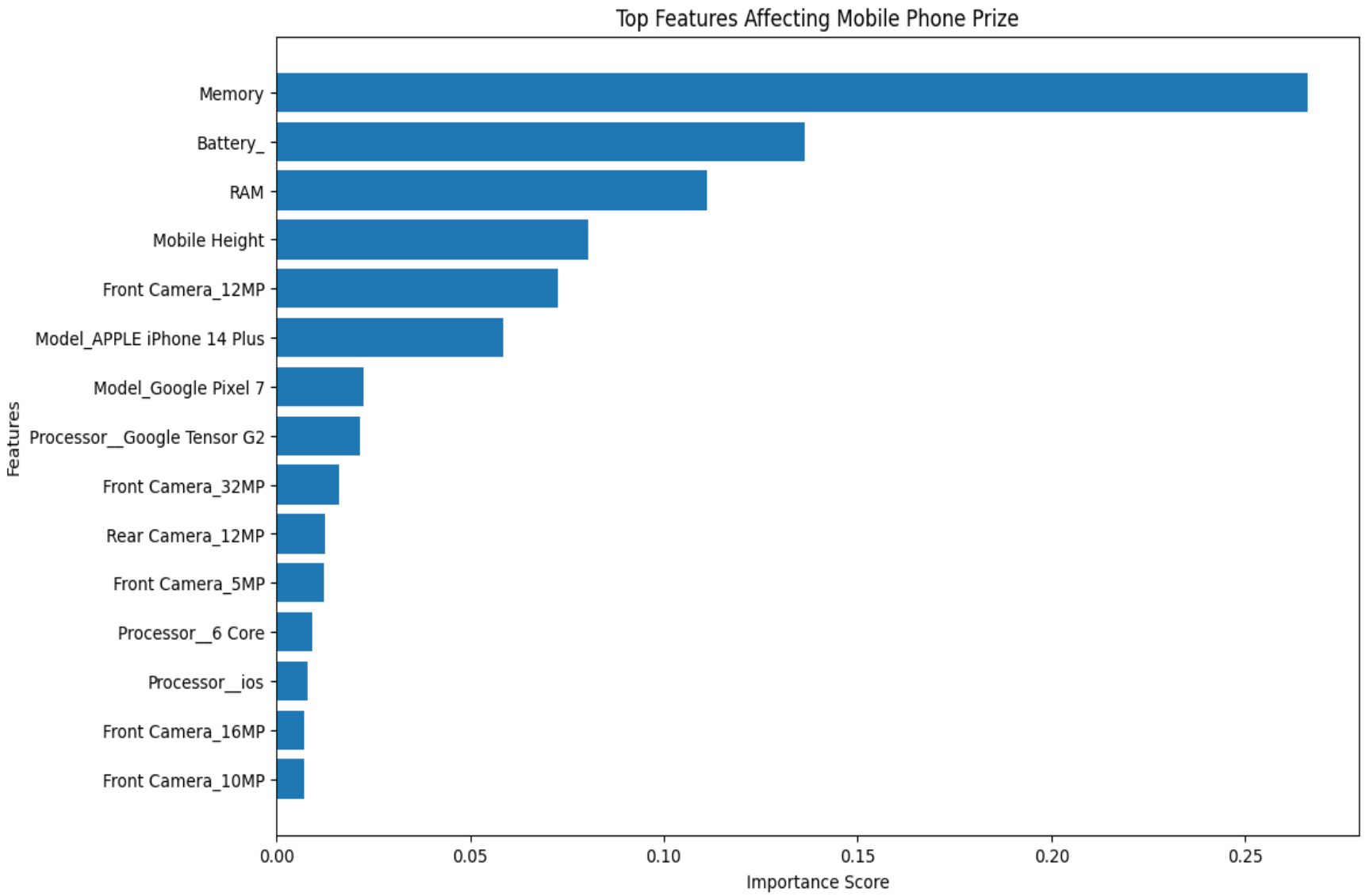
Top 20 Feature Importance Table

Feature	Importance
Memory	0.279671
Front Camera_12MP	0.232068
Battery_	0.123818
Mobile Height	0.061221
RAM	0.056203
Unnamed: 0	0.046171
Model_APPLE iPhone 14 Plus	0.022619
Processor__Google Tensor G2	0.016882
Front Camera_5MP	0.011131
Processor__ios	0.009997
Model_Google Pixel 7	0.009025
Front Camera_16MP	0.00642
Processor__Qualcomm Snapdragon 8+ Gen 1	0.006142
Front Camera_32MP	0.005784
Model_MOTOROLA Edge 30 Ultra	0.005382
Model_SAMSUNG Galaxy A34 5G	0.004813
Processor__Exynos 1380, Octa Core	0.004443
Rear Camera_50MP	0.004107
Processor__Dimensity 1080, Octa Core	0.003755
Colour_Phantom Black	0.00368

- Identifies which features contribute most to predicting price.
- Provides ranking of technical specifications.
- Most Important Features:- RAM; Memory / Storage; Battery Capacity; AI Camera / Lens Features; Device Dimensions.
- Feature importance directly reveals what customers are willing to pay more for.

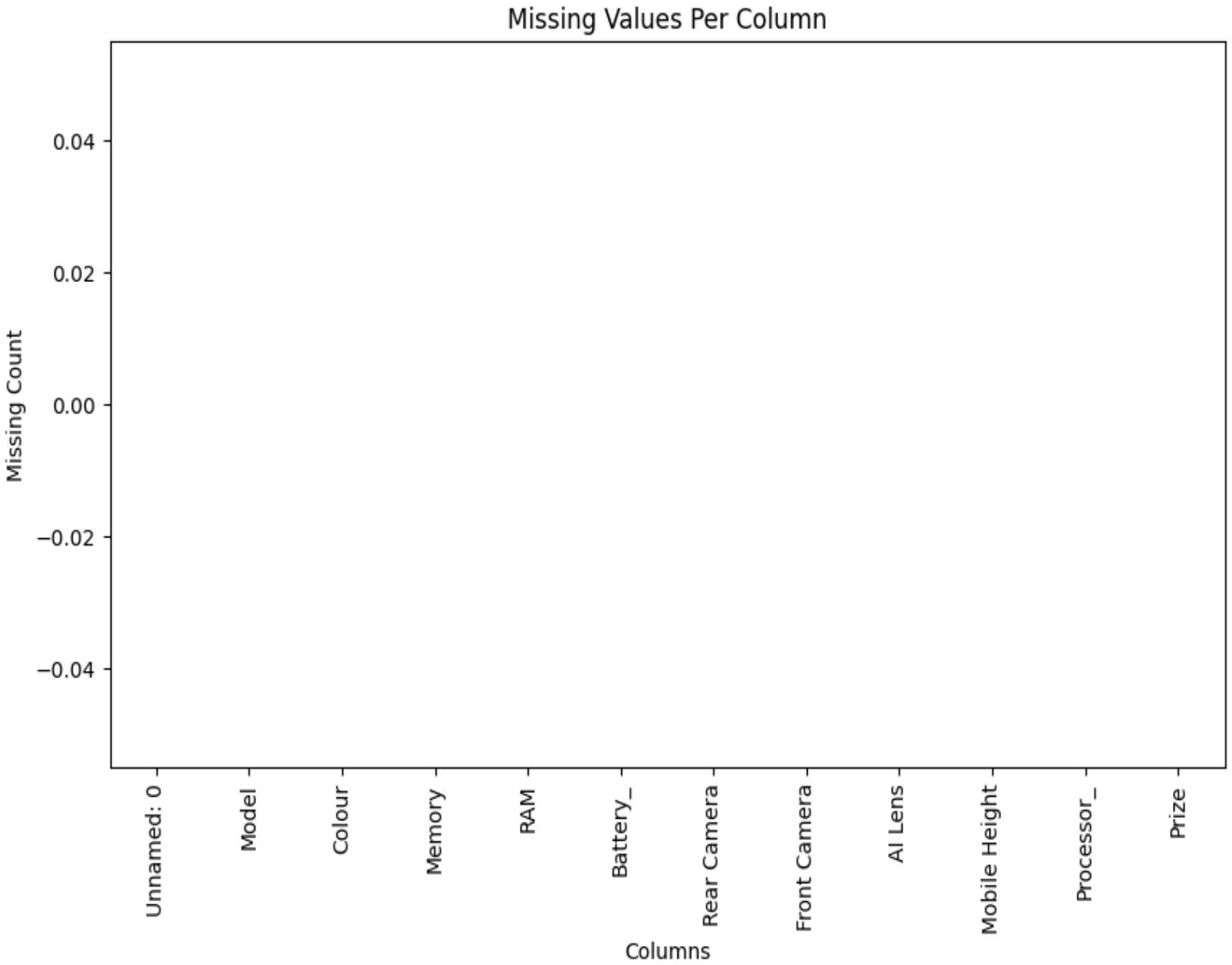
Table 3: Feature importance table of price prediction for mobile phones

Figure 3: Bar plots showing the top feature that affects the mobile phones prize



- This visually highlights top price-driving features.
- Makes it easy to communicate the strongest predictors.
- RAM and storage dominate pricing influence.
- Battery and camera provide secondary premium value.
- Product development should focus on top-ranked features to maximize profitability.

Figure 4: Showing the missing values per column for price prediction of mobile phones



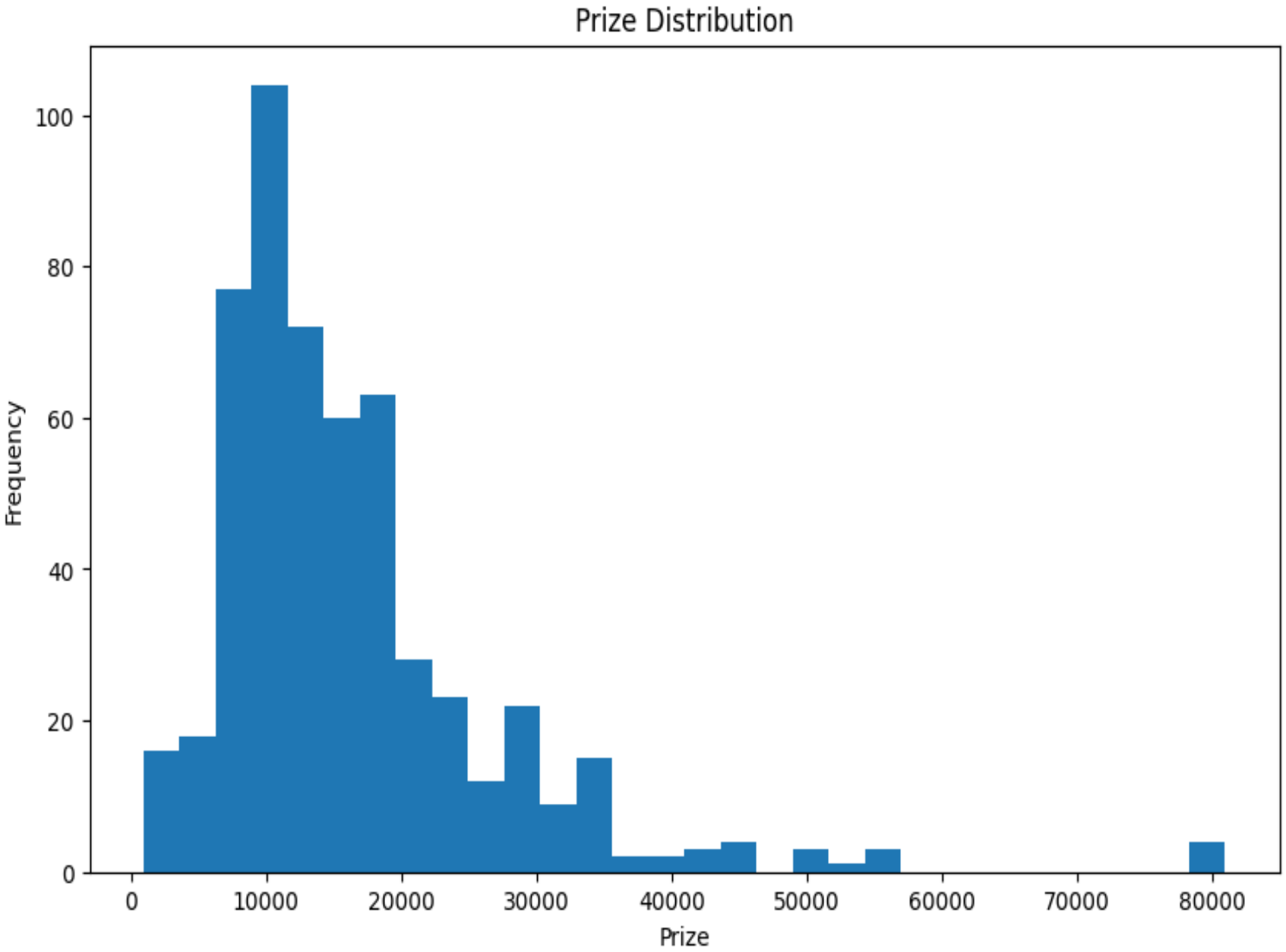
The missing value count:-

- ❖ Evaluates data completeness.
- ❖ Ensures prediction reliability.

The finding reveals that:-

- ❖ Minimal missing data improves model quality.
- ❖ Clean data leads to more accurate predictions.
- ❖ Strong data quality increases business confidence in model outcomes.

Figure 5: Histogram showing the prize distribution and its frequency among the mobile phones

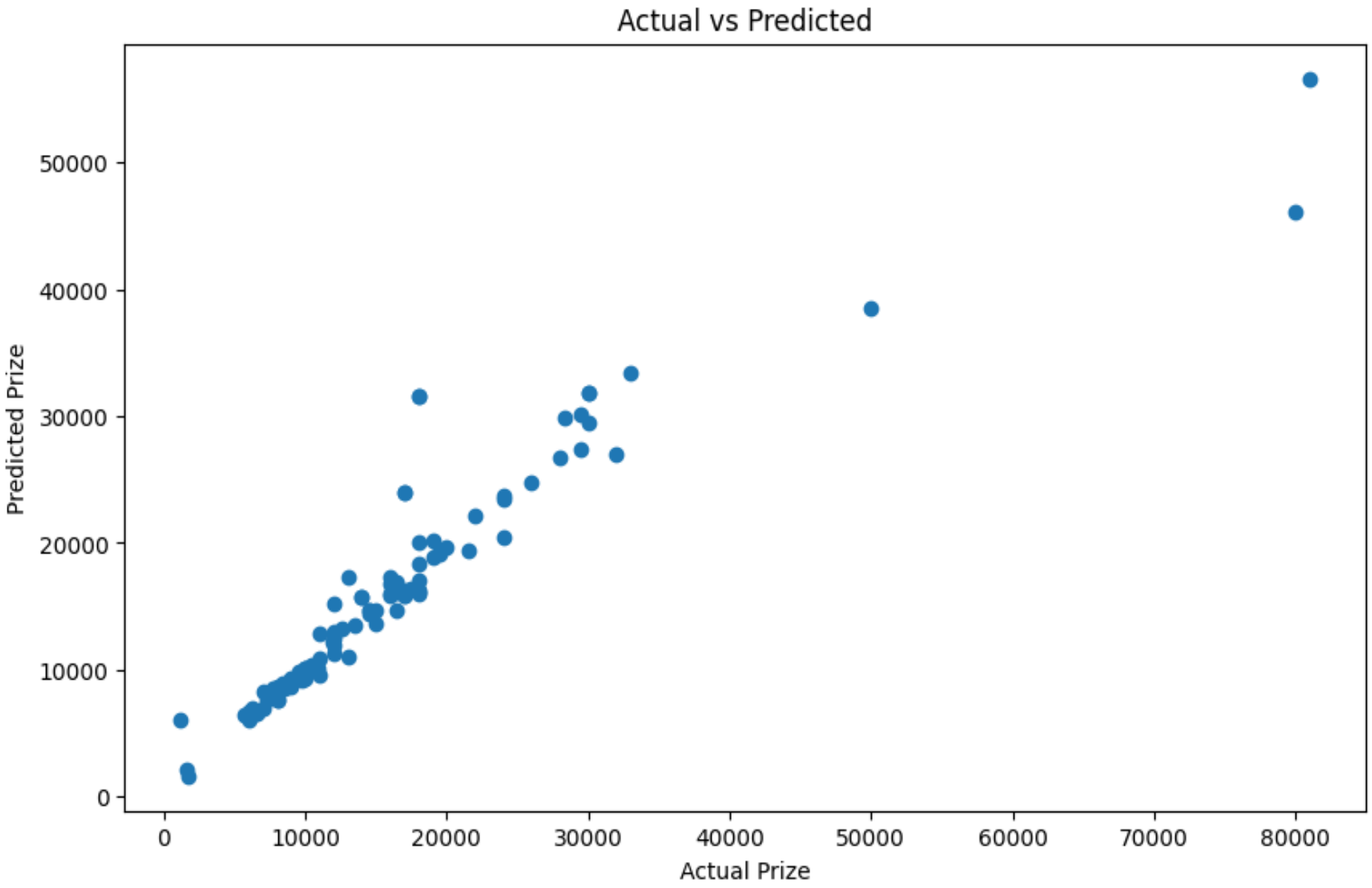


The figure shows distribution of mobile prices.

It reveals:-

- ❑ Customer purchasing segments.
- ❑ Market may be concentrated in mid-range pricing.
- ❑ Premium devices occupy smaller but higher-value segments.
- ❑ Helps companies identify dominant pricing categories.
- ❑ Useful for positioning products in competitive markets.

Figure 6: Scattered plots showing the actual versus predicted prize of the mobile phones



- The finding compares predicted prices with real prices.
- It indicates model performance quality.
- Points near diagonal line suggests accurate prediction.
- Strong alignment means high model reliability.
- The pricing model can effectively forecast mobile price based on specifications.

Figure 7: Scattered plots showing the residual plot of the mobile phones

- This plot shows prediction errors.
- Validates whether errors are random.
- Random residual distribution indicates model robustness.
- Absence of patterns suggests low bias.
- The model is statistically reliable for business deployment.

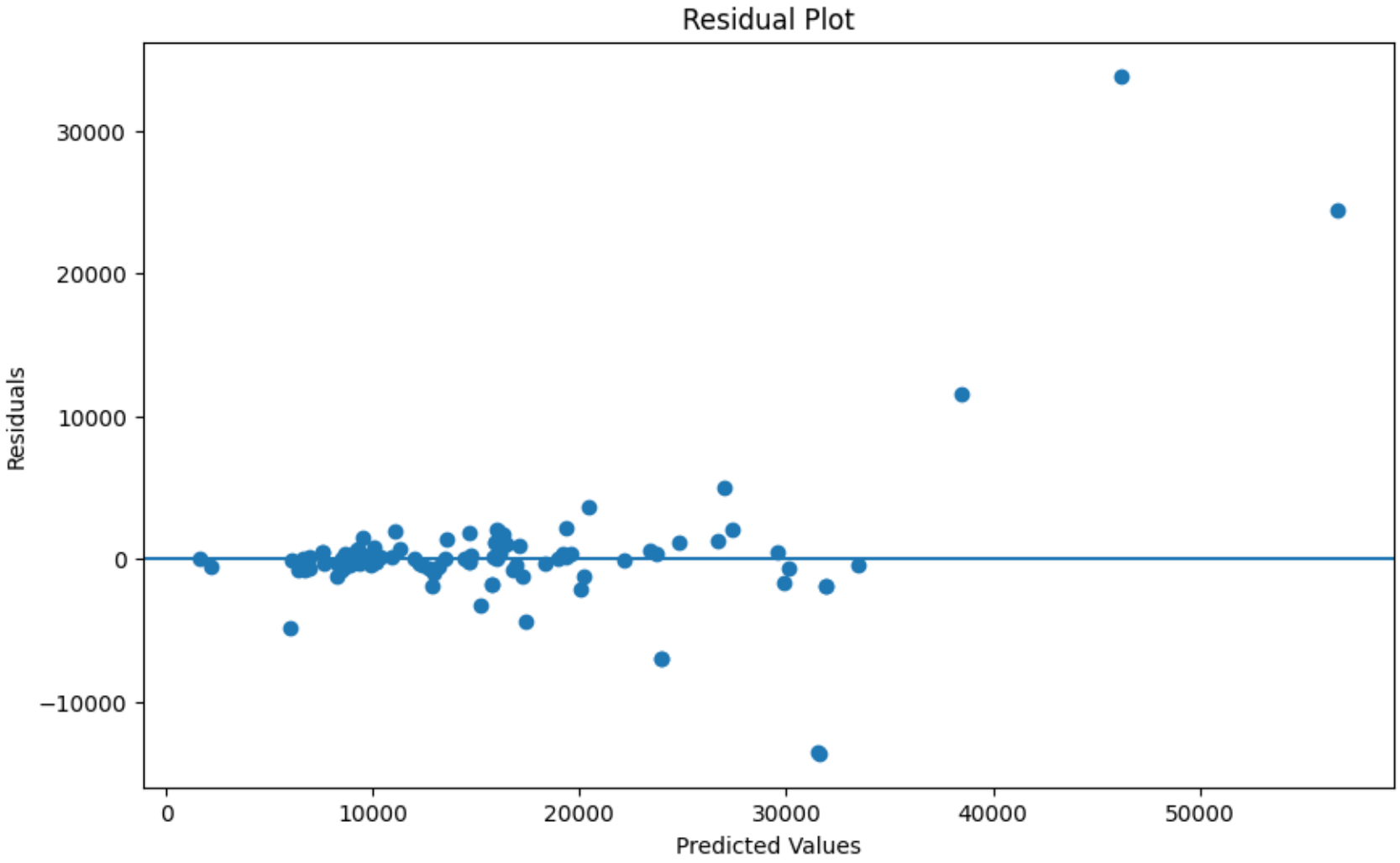
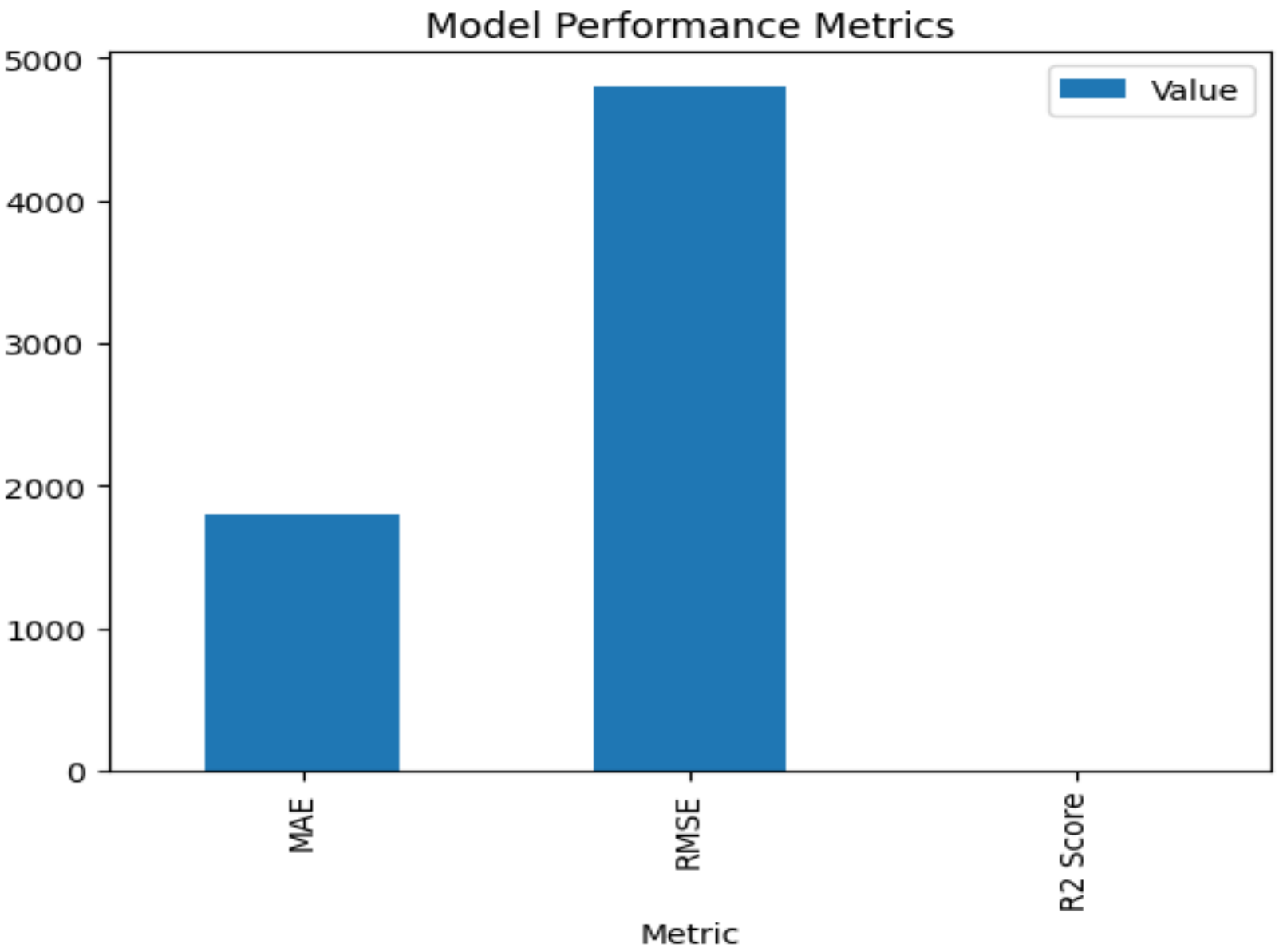


Figure 8: Bar plots showing model performance metrics of the mobile phones



The model performance metrics

- Compares machine learning model accuracy.
- Helps determine the best-performing prediction model.

These reveals

- One model likely outperforms others using lower error and higher accuracy.
- Supports selection of best algorithm.

Overall, we can deploy the highest-performing model for pricing automation.

Recommendations

- Prioritize High-Impact Features

  Focus future smartphone development on: Higher RAM options; Larger internal storage; Better battery capacity; AI camera enhancements; which means Higher perceived value = higher selling price.

- Introduce Tiered Product Strategy

  Budget Series (Standard battery, Moderate RAM, Basic camera); Mid-Range Series (Balanced specifications, Performance + affordability); Premium Series (High RAM, Large storage, Advanced camera systems, Premium design). Furthermore these improves market segmentation and customer targeting.

- Use Predictive Pricing for Market Advantage

  Deploy machine learning model to: Predict launch pricing; Benchmark competitor products; Reduce pricing uncertainty; Improve revenue forecasting. Notably, Data-driven pricing improves profit margins.

- Optimize R&D Spending

  Invest more in: Battery innovation; Performance upgrades; Camera technology; Reduce over-investment in low-impact cosmetic features. More specifically Higher ROI on product development.

- Strengthen Data-Driven Product Planning

  Use predictive insights to: Forecast customer demand; Plan inventory; Design feature bundles. Moreover, focus on operational efficiency.

However, the project demonstrates that mobile pricing can be scientifically predicted using technical specifications, allowing organizations to build smarter products, price competitively, and increase profitability through data-driven decision-making.